CO1-AT3

Descriptive Written Test

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1)

a) PEAS description.

component	Description
performance Measure	points awarded for clean rooms over time (+10), penalties for unnecessary movements (-1).
Environment	Two distinct locations, each capable of being in a state of clean or dirty.
Actuators	Wheels or motors to move Left and Right, vaccum mechanism to perform suck, and a mechanism to do nothing [rest].
sensors	Location sensor detects whether robot is in which room and Dirt sensor detects whether room is clean or Dirty.

b) Agent Type selection:

A model Based Reflex Agent is best suited to control this robot because

* By maintaining an internal state the robot can clean rooms based on the tracking records.

* once the model indicates ROOM A and ROOM B are clean, it knows it can DO NOthing rather than toggling between the rooms.

2)

Environment Property.	i) Playing chess against a Human	ii) Autonomous self-driving car in city traffic.	iii) solving a crossword puzzle alone
observability	Fully observable The entire board configuration piece position, legal moves are visible	Partially observable Blind spots, weather obstacles, and intent of other drivers limit complete state visibility	Fully observable The grid, clues and current filled-in letters are completely visible.
determinism	Deterministic An action is fully determined by chess rules	stochastic Traffic flow, sensor noise make outcome non-determined and probabilistic	Deterministic Filling in a letter chang the grid stat without unexpected randomnes
static VS Dynamic	static The board state does not change while agent deliberates its turn.	Dynamic The state changes continuosly while car deliberates	static. The puzzle state remai completely frozen wh solver thi
Discrete VS continuous	Discrete A finite set of distinct board location and valid turn based moves	continuous position and time vary continu-ously.	Discrete A fixed gr with finit set of discre cell coordi

3)

i) Initial state:

- Definition: $S_0 = (x_0, y_0)$ representing the robots starting coordinates in building grid, where $(x_0, y_0) \in$ Grid.
- Justification: Any search Algorithm requires a well defined starting point. specifying the initial coordinate establishes the root node of search tree.

ii) Goal state and Goal Test

- Goal state:

$S_g = (x_g, y_g)$, representing the targeted delivery point coordinates.

- Goal Test function:

$$\text{IsGoal}(S) = \begin{cases} \text{True}, & \text{if } S = (x_g, y_g) \\ \text{False}, & \text{otherwise.} \end{cases}$$

iii) Actions:

For a state $S = (x, y)$, the set of allowable actions is $A(S) \subseteq \{up, down, left, right\}$. An action is valid only if moving to adjacent cell does not hit an obstacle.

iv) Transition Model:

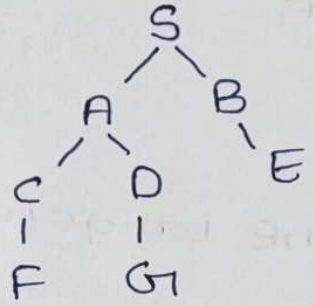
A function Result $(S, a) = S'$, where the executing action $a \in A(S)$ from state $S = (x, y)$ yields deterministic outcome.

- Result $((x, y), up) = (x, y+1)$
- Result $((x, y), down) = (x, y-1)$
- Result $((x, y), left) = (x-1, y)$
- Result $((x, y), right) = (x+1, y)$

3) Path cost:
$c(s, a, s') = 1$ for every step with total path cost $g(n) = \sum_{q=0}^{K-1} c(s_q, a_q, s_{q+1})$, where

K is total number of steps in path.

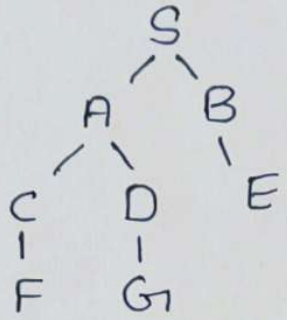
4) Breadth First search:



Step	Nodes Removed	Goal	children Added	Queue after step.
0	Start	-	-	[S]
1	S	NO	A, B	[A, B]
2	A	NO	C, D	[B, C, D]
3	B	NO	E	[C, D, E]
4	C	NO	F	[D, E, F]
5	D	NO	G1	[E, F, G1]
6	E	NO	None	[F, G1]
7	F	NO	None	[G1]
8	G1	yes	-	stops searching

Solution path = S → A → D → G1
path cost = 3

5) Depth First search :



step	Nodes popped	Goal	children pushed	stack after step.
0	start	-	-	[S]
1	S	NO	B, then A	[B, A]
2	A	NO	D, then C	[B, D, C]
3	C	NO	F	[B, D, F]
4	F	NO	None	[B, D]
5	D	NO	G1	[B, G1]
6	G1	yes	-	stops search

Solution path = S → A → D → G1.
Path cost = 3

BFS is more suitable Algorithm for unweighted graphs, even though DFS found the same path fastes.